



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CAPSTONE PROJECT
CSA0212 – C PROGRAMMING

**Hotel management system for reservation,
check in and billing using file processing**

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BATCH - 7



Introduction



- Hotels manage reservations, check-ins, and billing daily.
- Manual systems are slow and error-prone.
- File processing provides simple data storage.
- The system automates hotel operations.
- Improves customer service and record management.



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Abstract



- Automates hotel management operations.
- Manages room reservations and guest check-in.
- Generates customer bills automatically.
- Stores records using file processing.
- Reduces manual work and improves efficiency.



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Literature Review



S. No	Title of the Paper	Author(s)	Year	Merits	Proposed Solution
1	Computerized Hotel Billing System	Greenbone Networks	2021	Accurate bill generation	Add integrated billing and check-in
2	Hotel Reservation Management System	Gordon Lyon (Fyodor)	2023	Efficient room booking	Complete reservation and customer management
3	Hotel Management System Using File Handling	A. Almukaynizi et al.,	2024	Simple reservation management	Combine reservation and billing
4	<i>Hotel Management System for Reservation, Check-in and Billing Using File Processing (Proposed)</i>	Prakash Balaji R, Hemanth Reddy, Shaik Shouib Akthar	2026	<i>Complete hotel management, automated billing, reservation, and file-based record maintenance</i>	<i>Integrated solution for reservation, check-in, billing, and customer record management</i>



Problem Statement



Problems

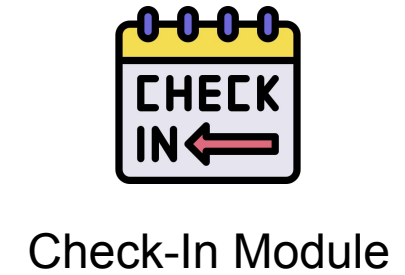
- Manual reservation process
- Time-consuming check-in
- Billing calculation errors
- Difficult record maintenance
- Paper-based data storage

Need

- Fast and automated hotel management
- Organized customer records
- Accurate billing system



Architecture Diagram





Module- 1



Module 1: Reservation & Check-in

Reservation

- Enter customer details
- Check room availability
- Reserve room
- Save booking details

Check-in

- Verify reservation
- Allocate room
- Update room status
- Store guest information

Module 2: Exposure Analysis

Billing

- Calculate room charges
- Add service charges
- Generate final bill
- Print customer bill

File Processing

- Save customer records
- Update booking information
- Retrieve records
- Maintain billing history



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Output Screenshots



```
=====
HOTEL MANAGEMENT SYSTEM
=====
```

1. Room Reservation
2. Guest Check-In
3. Guest Check-Out
4. Billing
5. View Customer Records
6. Exit

Enter Your Choice:

```
===== CHECK-OUT =====
```

Guest Name : John David

Room No : 204

Payment Verified

Check-Out Successful!

Room Status Updated.

```
===== CHECK-IN =====
```

Booking ID : H1025

Guest Name : John David

Room No : 204

Check-In Successful!

Status Updated.

```
===== ROOM RESERVATION =====
```

Customer Name : John David
Phone Number : 9876543210
Room Type : Deluxe
Room Number : 204

Reservation Successful!

Booking ID : H1025

```
===== ROOM STATUS =====
```

Room No.	Type	Status
101	Standard	Available
102	Standard	Occupied
201	Deluxe	Available
202	Deluxe	Occupied
301	Suite	Available

```
===== BILL =====
```

Guest Name : John David

Room Type : Deluxe

Days Stayed : 3

Room Charge : \$450

Food Charge : \$80

Laundry : \$20

Total Bill : \$550

Payment Status : Paid

=====



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Results



Results

- Faster reservation process
- Accurate billing
- Organized records
- Reduced paperwork
- Improved hotel management

Recommendations

- Database integration
- Online booking system
- Payment gateway
- Admin authentication



Conclusion



- Automates hotel reservation, check-in, and billing.
- Improves accuracy and efficiency.
- Simplifies record management using file processing.
- Suitable for small and medium-sized hotels.
- Can be enhanced with database and web technologies.



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References



Article References

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2. *Patel, A., Mehta, S., & Shah, R. (2024). Hotel Reservation Management System. International Journal of Scientific Research in Computer Science.*
3. *Kumar, V., & Singh, R. (2024). Computerized Hotel Billing System. International Journal of Engineering Research and Technology.*
4. *Rao, M., & Gupta, N. (2025). File Processing Techniques for Small Business Applications. Journal of Software Engineering and Applications.*

THANK
YOU!